FIH Mobile: Automating mobile handset manufacturing process with Google Cloud visual inspection products

Tell us your challenge. We're here to help.

Contact us (/contact)

About FIH Mobile

FIH Mobile Limited (Renamed to FIH Mobile Limited from Foxconn International Holdings in May 2013) was established in April 2002 and listed on Hong Kong Stock Exchange in February 2005 (Ticker: 2038.HK).

A subsidiary of Hon Hai Precision Industry and a leader in the worldwide mobile device industry, FIH Mobile offers vertically integrated, end-to-end design, development, and manufacturing services spanning handsets, mobile and wireless communication devices, and consumer electronics products. Moving into the internet era, FIH Mobile leveraged its core strengths in

FIH Mobile is res
business pressur
optimize efficien
Google Cloud vis
inspection produ
as AutoML Visior

automate and sta
defect inspection
smartphone man
operations.

Google Cloud results

- Inspects each component for defects in just 0.3 seconds
- Removes inconsistency by standardizing defect inspection across each station
- Enhances company's competitiveness in a rapidly changing market

Red
and
dec
defe
"esc
fron
10%
Goo

hardware and software to enter the 5G, AI, IoT and IoV (Internet of Vehicles) fields, building a full internet and mobile ecosystem.

FIH established a global footprint including design centers in Taiwan and China, manufacturing capacity in China, India, and Vietnam, and an after-market service center in the United States. In 2019, FIH reached $14.38 billion in revenue and acquired a 10.4% share of the global handset market. Moreover, until the end of 2019, FIH had accumulated 1,515 effective patents, comprising 560 software patents and 955 patents related to communication, mobile devices, and other hardware.

Industries: Technology

Location: Taiwan

Products: Google Cloud ... ML

Google Cloud (https://cloud.google.com/)

Cloud AutoML (https://cloud.google.com/automl/)

Tell us your challenge. We're here to help.

Contact us (/contact)

Consumers and businesses' demand for mobile devices, advanced driver assistance systems and in-vehicle infotainment systems, the Internet of Things (IoT), and 5G is powering growth at FIH Mobile (https://www.fihmb.com/). Traditionally, a leading manufacturer of mobile handsets, wireless communication devices, and consumer electronic products, the business is expanding into new markets to grasp the opportunities presented by changes in technology.

Established in 2002 and listed on the Hong Kong Stock Exchange in 2005, FIH Mobile has 71,000 employees and generated revenues of USD$14 billion in 2019. According to Eva Huang, Investor Relations, FIH Mobile has lines of business in mobile devices and the Internet of Things; an Internet of Vehicles division that builds and markets an "Advanced Driver Assistance System" and an in-vehicle infotainment system to vehicle manufacturers and Tier-1 suppliers; and a 5G/IoT integration solution.

"FIH Mobile offers vertically integrated, end-to-end design, development, and manufacturing services spanning handsets, mobile and wireless communication devices, and consumer electronics products," says Eva.

"The Advanced Driver Assistance System—through real-time tracking of information from sensors and by filtering out measurement noise and jitter—could automatically execute lane changes, emergency braking, adjustments to directions, and other actions—in different situations," she adds. "Meanwhile, the in-vehicle infotainment system features a voice assistant, facial recognition, navigation, and connected services including news, weather, and music, providing a multi-screen interactive experience that meets the needs of drivers and passengers.”

The system integration arms of the business provide software and hardware integration solutions for specific industries. For example, FIH Mobile offers a solution that embeds wireless sensors in cattle to enable farmers to remotely monitor their health conditions.

FIH Mobile’s mobile and wireless manufacturing operation spans product development and design, manufacturing and assembly, after-sales service, and repairs.

> “Humans inspected printed circuit boards (PCBs) in the pre-production phase of our original process. The defect escape rate—the number of unfound defects divided by the

number of all defective PCBs—was 40%. With AutoML Vision, we are able to reduce our defect escape rate to 10%, while the time to inspect each component decreased dramatically to 0.3 seconds, or 1.3 seconds once fixture movement time is taken into account. In addition, all our inspection stations apply the same inspection process, ensuring consistency and improving confidence.”

—Sabcat Shih, Senior Associate Manager, FIH Mobile

Becoming the best consumer electronics supplier

The business is a division of Hon Hai Precision Industry, the anchor company of Foxconn Technology Group. According to Eva, FIH Mobile aims to become the best global consumer electronics solution supplier, leveraging skills, expertise, and a physical presence that includes design centers in China and Taiwan, manufacturing facilities in China, India, and Vietnam, and an after-market service center in the United States.

FIH Mobile’s business is influenced by changes to downstream demand and upstream supply, with the coronavirus pandemic disrupting both areas. Consumers may be spending less in some markets, while fluctuations in supply chains can hamper availability of the materials needed to make the components of smartphones and other devices. Moreover, factors such as increasing competition, political tensions, and rising costs may also impact the company’s business.

> “We met the Google Cloud team and it provided information about how Google Cloud visual inspection products, including AutoML Vision, could help us.”
>
> *—Sabcat Shih, Senior Associate Manager, FIH Mobile*

Continuous improvement

FIH Mobile’s changing business profile and external factors such as escalating costs are increasing pressure on the organization to operate as efficiently as possible. The ability of machine learning to improve through experience presented a tremendous opportunity to improve the quality of its products, optimize processes, and reduce costs.

The manufacturer identified defect minimization as a potential “early win” for machine learning within its operations. During the smartphone manufacturing process, each printed circuit board (PCB) is inspected for defects that can compromise the quality of the final product delivered to a customer and sold or leased to a consumer.

“Humans inspected PCBs in the pre-production phase of our original process. The defect escape rate—the number of unfound defects divided by the number of all defective PCBs—was 40%,” says Sabcat Shih, Senior Associate Manager at FIH Mobile.

Clear information from Google Cloud

FIH Mobile decided to leverage its robust relationship with Google to identify a solution. “We met the Google Cloud team and it provided information about how Google Cloud Visual Inspection products, including AutoML Vision, could help us,” says Sabcat. “The team members were extremely helpful and provided clear business and technical information.”

AutoML Vision (/vision/automl/docs) enables teams to train custom machine learning models to classify images based on labels they define and build custom models to classify images at the edge to trigger actions based on local data.

Reviewing a set number of components per phone

FIH Mobile proceeded to build an automated system based on AutoML Vision and completed the deployment in April 2020. The business then began training machine learning models to support an image capture instrument that comprises two microscopes and an X-Y table that moves each PCB component-set horizontally through the defect detection process. The machine learning models support software that reviews and classifies the component in each image as “pass” or “fail.” The system reviews a set number of components in each phone.

“It’s super helpful applying these machine learning models in production. All we have to do is prepare the data, label it as accurate, and AutoML Vision does the rest for us.”

—Sabcat Shih, Senior Associate Manager, FIH Mobile

Reducing time and cost by decreasing defect escape rate to 10%

The business is achieving a considerable efficiency benefit from its decision. “With AutoML Vision, we are able to reduce our defect escape rate to 10%, while the time to inspect each component decreased dramatically to 0.3 seconds, or 1.3 seconds once fixture movement time is taken into account,” says Sabcat. “In addition, all our inspection stations apply the same inspection process, ensuring consistency and improving confidence.”

The Google Cloud (https://cloud.google.com/) team played a key role in helping FH Mobile deliver the AutoML Vision-powered solution effectively. “They helped us understand how to label data efficiently and maximize model performance,” adds Sabcat.

“It’s super helpful applying these machine learning models in production,” he concludes. “All we have to do is prepare the data, label it as accurate, and AutoML Vision does the rest for us.”

The business is now evaluating other Google visual inspection products that can provide a seamless pipeline from the image output to a machine learning model. “With Google Cloud visual inspection products, we are working towards a model where all we have to do is import our PCB images, select one image as a template, and draw an inspection area around each component's bounding box," explains Sabcat. "The product will undertake image alignment based on a template image, crop the components for us to label, and then train the machine learning model."